

YU HUO

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Shenzhen, Guangdong - 518172, China

EDUCATION

- **The Chinese University of Hong Kong**  2026 - Incoming
Master of Science — Artificial Intelligence (Offer Received) Hong Kong, China
- **The Chinese University of Hong Kong (Shenzhen)**  Sep 2022 - Present
Bachelor of Engineering — Electronic and Computer Engineering (Minor in Artificial Intelligence) Shenzhen, China
 - MGPA: [3.60]/4.00, Major Rank: Top [6]
 - Core Courses: Natural Language Processing, Deep Learning and Applications, Advanced Machine Learning, Machine Learning, Fundamentals of Artificial Intelligence, Design and Analysis of Algorithms, Modern Optimization, Optimization, Linear Algebra, Data Structures, Database Systems, Software Engineering, Stochastic Processes, Discrete Mathematics, Probability and Statistics I & II, Calculus I & II

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, A=ARXIV PREPRINT, T=THESIS

- [C.1] **Huo, Y., Zhang, S., Zeng, K., Liu, H., Lee, O., Chen, J., Lu, Y., Guo, Y., Liang, Y., and Tang, X. (2026). Shape of Thought: Progressive Object Assembly via Visual Chain-of-Thought.** In *Forty-Third International Conference on Machine Learning (ICML)*, 2026.  
- [C.2] **Huo, Y., Zhang, S., Zeng, K., Lu, Y., Yang, C., Guo, Y., and Tang, X. (2026). RepoShapley: Shapley-Enhanced Context Filtering for Repository-Level Code Completion.** In *The 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.  
- [C.3] **Huo, Y., Li, Y., and Tang, X. (2025). FedGF: Layer-Wise Federated Learning with Group Fairness Guarantees.** In *Proceedings of the International Conference on Intelligent Computing (ICIC)*, 2025.  
- [S.1] **Zeng, K., Huo, Y., Zhang, S., Ye, Z., Zhuo, Y., Liu, H., Lu, Y., Wen, J., and Tang, X. (2026). Group of Skills: Group-Structured Skill Retrieval for Agent Skill Libraries.** Manuscript submitted to *Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2026.  
- [S.2] **Yan, H., Huo, Y., Lu, M., Ou, W., Shi, X., Shi, R., and Tang, X. (2025). Optimal Boost Design for Auto-bidding Mechanism with Publisher Quality Constraints.** In *arXiv preprint arXiv:2508.08772*, 2025. Manuscript submitted to *ACM Transactions on Information Systems (TOIS)*.  
- [S.3] **Guo, Y., Chen, S., Lu, Y., Lin, J., Xu, Z., Lin, J., Zhang, S., Yang, C., Li, J., Li, Y., Huo, Y., and Wang, R. (2026). From Brewing to Resolution: Tracing the Internal Lifecycle of Code Reasoning in LLMs.** Manuscript submitted to *3rd Conference on Language Modeling (COLM)*, 2026. 
- [A.1] **Lu, Y., Guo, Y., Xu, Z., Zhang, S., Huo, Y., Chen, S., Wu, S., Zhu, C., and Wang, R. (2026). Zero-Forgetting Class-Incremental Segmentation via Dual-Phase Cognitive Cascades.** In *arXiv preprint arXiv:2603.13874*, 2026. 
- [P.1] **Tencent Technology (Shenzhen) Co., Ltd. and The Chinese University of Hong Kong, Shenzhen. (2025). Method for Determining a Boost Factor and Related Apparatus.** Chinese invention patent application No. 202511388691.1, filed Sep. 25, 2025. Related patent application for q-Boost.

RESEARCH EXPERIENCE

- **First Author, "Shape of Thought: Progressive Object Assembly via Visual Chain-of-Thought"** Jan 2026
Guangdong Provincial Key Laboratory of Future Networks of Intelligence — Accepted, ICML 2026  
 - Proposed Shape-of-Thought (SoT), a visual Chain-of-Thought framework that generates interleaved textual plans and rendered intermediate states for progressive object assembly.
 - Built the SoT-26K dataset by converting part-based CAD assets into step-aligned multimodal traces via automated hierarchy decomposition and Blender rendering, enabling process-supervised training.
 - Developed T2S-CompBench to evaluate structural integrity and trace faithfulness, combining structure metrics with process-oriented assessments.
 - Fine-tuned a unified multimodal autoregressive model (Bagel-7B variant) on interleaved traces, achieving substantial gains on component numeracy and structural topology over direct generation baselines.
- **First Author, "RepoShapley: Shapley-Enhanced Context Filtering for Repository-Level Code Completion"** Jan 2026
Guangdong Provincial Key Laboratory of Future Networks of Intelligence — Accepted, ACL 2026  
 - Proposed RepoShapley, a Shapley-enhanced context filtering approach to improve repository-level code completion by selecting higher-value contextual chunks.

- Designed the context contribution estimation pipeline and integrated it into a repository-aware completion workflow to reduce irrelevant context and improve completion quality.
- Conducted comprehensive evaluations on repository-level completion settings and summarized findings for paper writing and submission preparation.
- **Co-first Author, "Optimal Boost Design for Auto-bidding Mechanism with Publisher Quality Constraints"** *Mar 2025*
Tencent Online Advertising BG, Shenzhen, Guangdong, China 
 - Developed a novel Boosting algorithm, incorporating publisher quality constraints, which improved welfare efficiency by 2-6% compared to existing methods.
 - Conducted theoretical analysis and designed an efficient three-party auction framework involving advertisers, publishers, and the platform, balancing commercial and quality metrics.
- **First Author, "FedGF: Layer-wise Federated Learning with Group Fairness Guarantees"** *Mar 2025*
The Shenzhen Institute of Artificial Intelligence and Robotics for Society 
 - Led the research project on Federated Learning (FL), proposing a novel method (FedGF) to incorporate both client-level and group-level fairness in federated learning systems.
 - Conducted extensive experiments on benchmark datasets (FMNIST, CIFAR-10) using multi-objective optimization to ensure fairness without sacrificing model accuracy, achieving 78% reduction in group accuracy gaps.
 - Developed and implemented the FedGF approach using advanced machine learning algorithms and statistical methods to resolve gradient conflicts at both the model and layer levels.

INTERNSHIP EXPERIENCE

- **Huawei Technologies Co., Ltd.**  *Jul 2025 - Oct 2025*
AI Algorithm Engineer *Shenzhen, China*
 - Assisted in improving the performance of CodeMate, an AI-powered code completion tool, by fine-tuning large models using preprocessed code data and SFT data extraction, resulting in a 5-10% improvement in tool performance.
 - Applied high-quality corpus filtering and integrated RAG-based techniques, leading to a significant increase in the adoption rate of the tool.
- **Shenzhen Research Institute of Big Data**  *Oct 2025 - Present*
Research Assistant *Shenzhen, China*
 - Developed ICHAgent, a ReAct-style clinical decision-support agent for intracerebral hemorrhage GOS prediction, coordinating clinical-table lookup, AutoGluon ML prediction, nnUNet CT feature extraction, medical search, and structured reasoning traces.
 - Built Case RAG and medical rule-KG support for historical ICH comparison and knowledge-grounded reasoning, combining weighted KNN retrieval with single-factor GOS distribution-shift rules exported as structured JSON.
 - Implemented evidence-gated prompting and resumable evaluation with top-2 GOS candidates, confidence scores, top-1/top-2 accuracy, classification reports, and confusion matrices.

SKILLS

- **Programming:** Python, C/C++, Java, JavaScript, TypeScript, SQL, HTML/CSS, $\text{\LaTeX}$
- **Tools:** PyTorch, Hugging Face, Transformers, Datasets, FAISS, scikit-learn, NumPy, Pandas, Git, Docker
- **Language:** Chinese (native), English (fluent)

HONORS AND AWARDS

- **Undergraduate Research Award** *Mar 2025 - Present*
The School of Science and Engineering, The Chinese University of Hong Kong, Shenzhen 
 - Recognized for outstanding academic performance and the potential to produce high-quality academic papers and research output.
- **Dean's List** *Aug 2022 - Aug 2024*
The School of Science and Engineering, The Chinese University of Hong Kong, Shenzhen 
 - Recognized for achieving high academic standards and maintaining exceptional performance throughout the year.
- **Meritorious Winner** *Jan 2025*
Mathematical Contest in Modeling (MCM), Consortium for Mathematics and its Applications (COMAP) 
 - Applied deep reinforcement learning and dynamic programming to optimize tourism, environment, and social welfare, resulting in a 40% increase in tourism revenue and a 25% reduction in carbon emissions.
- **Provincial First Prize** *Sep 2025*
China Undergraduate Mathematical Contest in Modeling (CUMCM) 
 - Built geometric occlusion criteria and formulated optimization models under multi-UAV / multi-decoy / multi-target settings, solved via the Cross-Entropy Method (CEM) and progressive refinement strategies.